



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 1I: Lab 1 Intro

Prior to Starting Lab 1

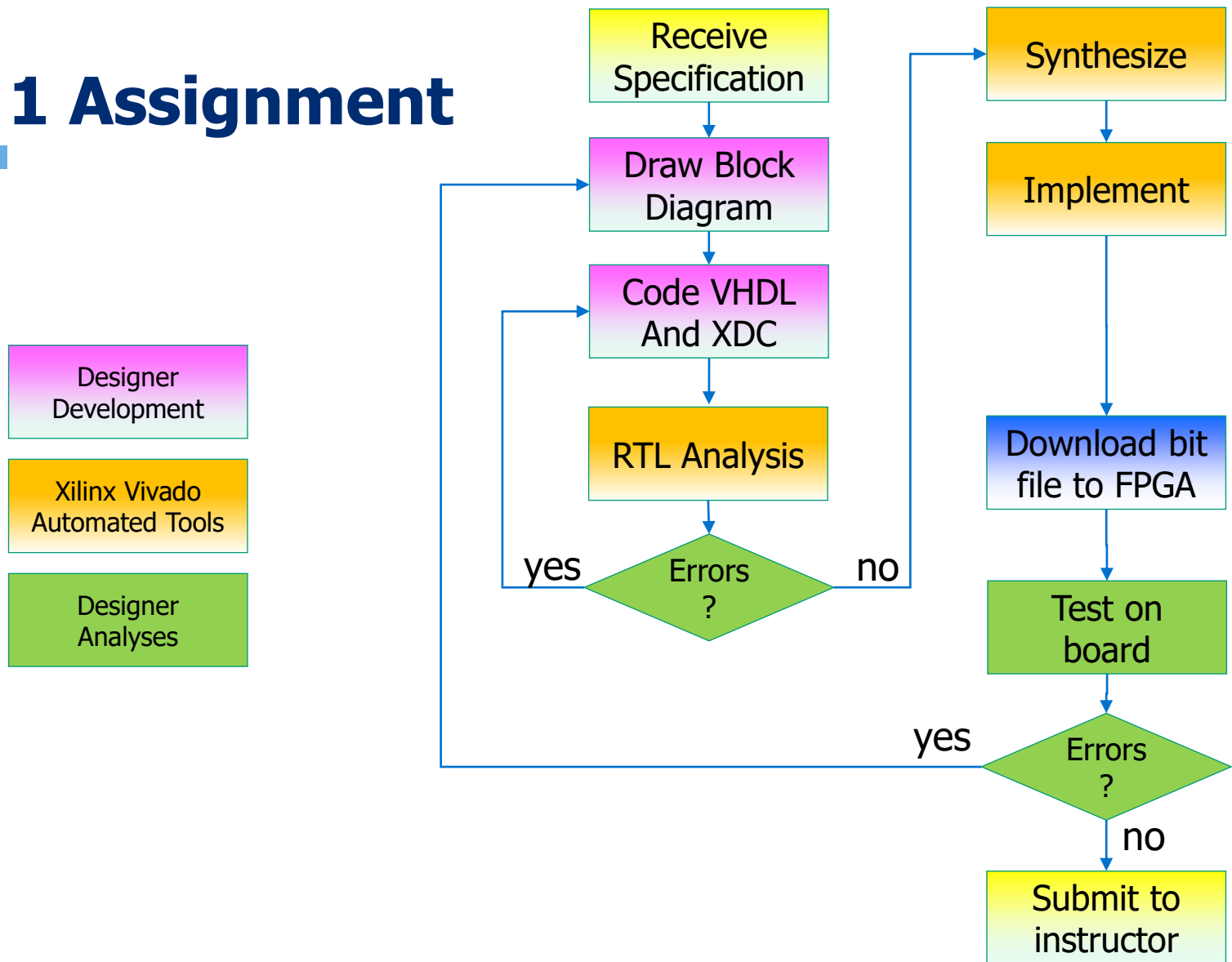
- Install Vivado Standard:
 - <https://www.xilinx.com/support/download.html>
 - Create a Xilinx Account using your JH email to download and install
- Follow Nexy's/Vivado Tutorial
 - Located on Canvas

Lab 1 Overview



- First part is setting up Vivado and running through the process of importing VHDL to programming the FPGA (Tutorial)
- Second part is a lab using basic combinational logic to control the 7-segment display's on the development board

Lab 1 Assignment



General FPGA Assignments

- Minimum requirements for all designs:
 - VHDL files describing the function of the design
- The Vivado Constraints file (.xdc)
 - Tells the tools (Xilinx Vivado) how to map the FPGA I/O to the board peripherals
- NOTE: You CAN synthesis without a XDC file
 - Design will not work properly when programmed

Words of Advice



- Always start a new project for each lab assignment. DO NOT COPY previous project data as this can cause problems
- Always make sure to have a constraints file (.XDC)
- Constantly save your work
- Think in terms of hardware components not software (except picoblaze lab)